Light Fidelity, or LiFi, is a wireless optical networking technology that offers remote transmission of data using LED light. In effect, LED bulbs will bring not only light but wireless connection at the same time. This is done by coaxing that innocuous-looking orb to emit not only white light that fills your room but also light pulses that are invisible to the human eye. Those pulses can encode bits of data and convey them to receivers within the line of sight, endowed with the capacity to interpret them. This latter operation is similar to the decoding of Morse, only at a much, much faster clip. With LiFi, transmission speeds can exceed 100 Gbps, making it over 14 times faster than WiGig (60 GHz WiFi), reckoned as the world's fastest WiFi. LiFi also provides better latency, performance, accessibility and security than WiFi, and also has no radio interference issues.

WHY IT IS A GAME CHANGER

The technology is not just cheaper but more efficient and can work without electronic devices such as routers and modems. It can also work in areas vulnerable to electromagnetic interference, such as hospitals and airplane cabins. With the ability to transmit data at speeds of up to 224 gigabits per second, LiFi could be the answer to the endless demand for bandwidth in a data-driven world. LiFi's vastly greater bandwidth also makes it ideal for use in densely populated urban areas. In fact, street lamps could soon become data signalling devices, providing information to pedestrians as well as vehicles, revolutionising the functionality of city infrastructure. While currently concentrated in industrial settings, LiFi will soon make its way into smart homes and become an integral part of building automation.

LiFi technology is already providing high-speed internet access for educational institutions in far-flung Ladakh and in the Akrund and Navanagar villages in the Aravalli district of Gujarat that have evolved as smart villages. Schools, hospitals, post offices and other government offices in these villages are LiFi-enabled. It is also meeting the internet broadband needs of the Border Security Force and tourists at Nadabet, one of the most secluded points on the Indo-Pak border in Gujarat.



"The launch of LiFi and innovative offerings in optical wireless technologies come at a time when very high speed internet connectivity is much needed. It promises to address issues head-on"

–HARDIK SONI
Co-founder and CTO, Nav Wireless Technologies

THE CHANGE-MAKERS

NAV WIRELESS TECHNOLOGIES

- The Ahmedabad-based firm is the pioneer of LiFi in India, helping Akrund and Navanagar in the Aravalli district of Gujarat evolve as smart villages, getting fast internet connections over existing electric power lines
- Meeting the broadband needs of the BSF and tourists at Nadabet on the Indo-Pak border in Gujarat; has set up a LiFi network for the Students Educational and Cultural Movement of Ladakh

WIPRO LIGHTING

- The Wipro Enterprises subsidiary is partnering with Scotland-baseb pureLiFi, global leaders in LiFi technology, to develop applications for wireless communication and the lighting market in Asia

VELMENNI

- The Delhi-based start-up has created a LiFi-based mesh network solution that allows transmission of data across long distances. It is working with the defence ministry for a customised LiFi solution in indoor/outdoor scenarios

WHAT INDIA NEEDS TO MASTER

LiFi could potentially turn every light source in a city into a data transmission point, revolutionising the way traffic, monitoring of utilities and delivery of public services in smart cities is managed. The challenge is that it requires a direct line of sight between the transmitter and receiver. Concerns over the limited range of light, its inability to penetrate through walls, is good from a security perspective, but hampers its application in certain environments. LiFi is good for indoor purposes in an enclosed area with minimum ambient light as external light may mess with the LiFi signals. This may hamper the technology's growth. Right now, LiFi is coming up as a ubiquitous systems technology consisting of application-specific combinations of light transmitters, light receivers including solar cells, computational algorithms and networking capabilities. The initial costs of setting up LiFi networks can also be a constraint. ■